



Winding Tree

Summer 2018 release — the technical description



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After reading the first part of this Summer release series, you should have a clear understanding of what the platform is capable of right now. In today's post, I will show you how everything is glued together.

I have split the post into three segments: Data structure, Infrastructure and Tooling. They are not really tightly connected, so feel free to skip to a section of your interest.

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Data structures

This is a hard one. Even after doing a lot of research, we're still unsure if we came up with a good design. Take for example price per room per night: It might depend on the number of people, on their age, the part of the year, the length of stay... And every other system that might use a slightly different approach.

On data shapes and formats

After a few weeks of tedious research of many existing APIs and standards, we are ready to show to the world how the...

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We are very open to all of your suggestions on how to improve it and we know we have totally not covered everything. Even the availability schema we came up with will probably change when we actually implement it in the coming weeks.

On the other hand, there will never be a standard that would fit all use cases. So at some point we have to draw a line and figure out if we want to make it easy for existing systems (using OTA or HTNG) to integrate, or if we want to make it easy to work with modern dynamic pricing. Tough choices.

So we are publishing what we think will work for a lot of hotels. But we encourage you to give us as much feedback as possible to improve the current state. A great way would be to collect common use cases and document how they can or cannot be recorded in the proposed Winding Tree data structure.

Finally, for future reference: The data structure might (and probably will) change. Since we are still in a phase of demos and testing deployments, it shouldn't be a huge problem. But be aware of this if you are working on Winding Tree integration. Once we are confident that the hotel data structure is somewhat stable, we will let you know.

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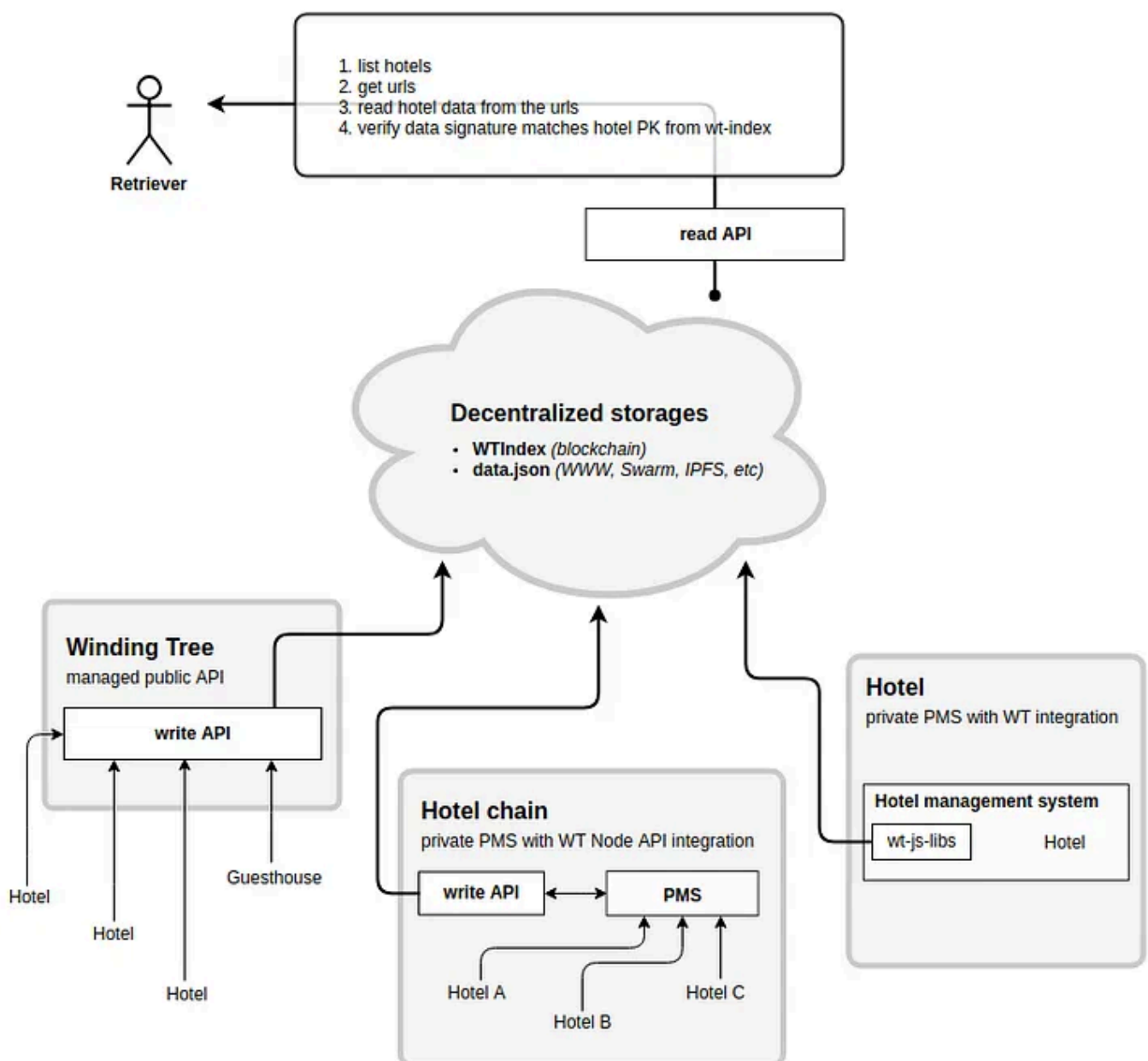
Infrastructure

Since the storage research and the first architecture announcement, we were confident that we could pull this off. The initial idea was to support any storage you would like, and we actually can. The only thing for you is to pick the right adapter. Or, you know, write one.

Our initial goal was to bypass slow transaction times with storing data off-chain, but we still needed to keep a root URI in a smart contract. And that's a speed-limiting factor if the storage technology cannot mutate a resource while keeping its URI stable over time (because you would still need to do a transaction with each update). That's why we allowed the use of good old HTTP.

With the POC 3 release of Swarm, our hopes went up a lot. And when we were able to hack on the MRUs (Mutability Resource Updates) in ETHBerlin, it was a game changer. Once Swarm is stable enough we should be able to easily use it for hosting all of the platform's data.

WT architecture overview



While experimenting with Swarm-based solution, we also came up with an easy-to-deploy Swarm node connected to the same list of nodes as swarm-gateways.net is.

This general concept of having off-chain data linked from smart contracts is pretty cool and very re-usable. We plan to use this pattern in the future for more parts of the platform.

Another benefit of this solution is that your data is not hosted on any proprietary platform. Winding Tree does not own the data. Winding Tree does not host the data. Everything on Swarm is distributed and decentralized in a similar way to Ethereum blockchain itself.

The only actual part missing here is an efficient way of keeping tabs on data changes. And this kind of data can change *a lot*. So it's probably unfeasible to poll all the time for all of the data documents you are interested in. Imagine that an online travel agency would be constantly checking on thousands of hotels. That's a no-no.

So in September, we kicked off a new piece of infrastructure called wt-updates-api which should work as a publish-subscribe solution for getting notifications about changed data. We believe that this will significantly improve the overall speed of data distribution to all of the applications built on top of the Winding Tree platform.

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Tooling

To make the use of the infrastructure as simple as possible, we have developed multiple tools (for now only in JavaScript) that should make your life easier if you want to work with the Winding Tree platform. These are:

- [wt-js-libs](#) — NPM module that you can use to directly interact with Winding Tree’s smart contracts and off-chain stored data.
- [wt-read-api](#) — REST API that wraps around both on-chain and off-chain data and allows you to read all of the data in a web-friendly way.
- [wt-write-api](#) — REST API that can help you to integrate your current system into the platform and allows you to create, update and delete hotel data.

For demonstration purposes, we are also running two separate instances of the Winding Tree platform on [Ropsten](#) testnet, you can check the details on [GitHub](#) or browse the data in our [Explorer: demo](#) and [playground](#). All of these are running only when there’s some traffic, so it might take a little time for them to warm up. You can find some example code in [our developer guides](#).

The wt-write-api should never be run in the wild like this for production deployments, though. By storing client keyfiles (even if password-protected) locally in the instance, we are introducing a potential attack vector that could lead to losing some funds. *If you will be experimenting with the API, use a dedicated keyfile that is not used anywhere else.* This API is actually meant to run in a local secure network inaccessible from the internet. In a public setup like this, the transactions should be signed by a totally separate component. We might work on that at some point in the future.

The Hotel Explorer is meant as an example application that shows how easy it is to use data from Winding Tree in a conventional web app (it’s a React + Redux app getting the data from the appropriate wt-read-api).

If you’d like to know more technical details, check [our wiki](#) for now, we plan to build a full-featured standalone developer portal in the near future.

Once we have all of the infrastructure (notifications about data changes) and data ready (remember, availability is still missing), we will work on a production-ready version of *Booking* and *Payment*.

In the third part of the Summer release announcement, I will try to describe how we develop all of this and how you can track our work even when there are no blogposts.

Thank you for your support and patience! If you have any questions, join our Telegram, technical discussion should go to our Google Group. If you want to contribute, a good place to start is GitHub or our upcoming hackathon!

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